

Emily O'Neal

831-359-7241 - emily.anna.oneal@gmail.com - emsoneal.github.io

EXPERIENCE

Industrial Light & Magic - Gameplay Engineer

2022 - Current

Millennium Falcon: Smugglers Run – Attraction Update

- Implemented core gameplay mechanics and served as ILM's technical point of contact for WDI and DPEP on multi-application UI pipelines and in-cabin hardware feedback.
- Built an Unreal Engine state system bridging telemetry with Unity UI states, supporting flexible gameplay behavior and synchronized media playback.
- Developed in-cabin debugging and tuning tools for engineering and creative teams to iterate on attraction behavior.
- Supported cross-disciplinary debugging during overnight testing, coordinating triage between software and hardware teams under tight deadlines.
- Investigated and reverse-engineered legacy attraction systems to support testing and create targeted scenarios and debugging workflows for engineering and design teams.

Beyond Victory — Mixed Reality Experience

- Led engineering for Create Mode, a mixed-reality sandbox for interactive character manipulation. Prototyped mixed-reality interaction systems, translating experimental concepts into shipped features.
- Designed and implemented a Control Rig–driven procedural posing system in Unreal Engine, enabling technical animators to author skeletal behaviors and gameplay interactions without engineering support.
- Architected an automated UMG asset pipeline for independent artist content management.

PlayStation - Software Engineer

2020 - 2022

- Architected, designed, and implemented a third-party integrated service bridging users and games, with a focus on web- and mobile-based remote control of the PS5 gaming console. Supported PlayStation's 20+ web microservices.
- Collaborated with a seven-person team to develop a platform that reimaged how people play games virtually. Contributed to game design. Programmed graphic effects and animations. Implemented multiplayer functionality. Integrated game into the larger platform.

Amotion Games - Co-Founder

2019 - 2022

- Built a cross-functional team to develop a platform delivering healthy gaming to households via a web-based motion tracking system using standard computer hardware.
- Developed a proprietary environment in Unreal, optimizing for computational efficiency to bring the motion-tracking experience to web browsers.

Mixed Reality Lab (Cal Poly) - Undergraduate Researcher

2019 - 2020

- Developed real-time projection mapping systems that transformed humans into interactive canvases for visual effects.
- Directed interactive dance performances with Kinect and projection systems, expanding storytelling possibilities in dance.

SKILLS

Engines & Graphics: Unreal Engine, Unity, WebGL, OpenGL, PlayCanvas

Languages: C++, C#, Python, JavaScript, Java, GLSL

Technologies: Mixed reality, VR Interaction, Motion Tracking, Spatial Computing, Real-Time Rendering

Tools & Hardware: Perforce, Git, Meta Quest, Kinect

EDUCATION

Cal Poly San Luis Obispo, CA

BS: Computer Science

Concentration: Interactive Entertainment **Minor:** Dance

LEADERSHIP

Dance Instructor: Taught social dance classes and emphasized movement, rhythm, and collaborative learning

Founder: Created hackathons, organizing 400+ students and securing \$70K+ in funding to expand accessible technical education across regional events

WISH Officer: Created more accessible spaces for underrepresented students to expand their technical skills

Curriculum Developer & Educator: Re-shaped curriculum to project-driven learning, then taught for K-12 students